

Do Ensemble Models Always Win? A Comparative Machine Learning Evaluation for Financial Time-Series Prediction

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Abstract

Equal-weighted ensembling is often treated as a default upgrade over individual machine learning models. We test this assumption on one-day-ahead stock-price prediction across five sector-diversified U.S. assets (2015–2024) by comparing Random Forest (RF), Support Vector Regression (SVR), Long Short-Term Memory (LSTM), and their equal-weighted average under identical features, preprocessing, and chronological splits. Pooled across assets, LSTM attains the lowest test-set RMSE (17.70), SVR is statistically tied with LSTM under the Diebold and Mariano test ($p = 0.494$), the equal-weighted ensemble is third (18.13), and RF is last (20.33); SVR beats the ensemble at $p = 0.001$ (Bonferroni-significant) while LSTM beats it at $p = 0.035$ (marginal under family-wise correction). The ensemble inherits intermediate accuracy and a positive bias between those of its members. The bias and variance decomposition predicts this ranking when constituent models have unequal accuracy, correlated errors, and heterogeneous bias; from the reported error standard deviations the implied average pairwise error correlation is approximately 0.94, and the heterogeneous-bias and unequal-accuracy conditions are visible directly. Equal-weighted ensembling does not automatically improve on careful individual model selection in this regime; validating its underlying assumptions on validation data is the methodological lesson.

CCS Concepts

• **Computing methodologies** → **Machine learning**: *Neural networks*; *Supervised learning by regression*.

Keywords

comparative evaluation, ensemble learning, financial time-series, LSTM, random forest, support vector regression

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1 Introduction

Aggregating predictions from several heterogeneous models is widely assumed to outperform any individual model. The bias and variance decomposition makes the assumption explicit: averaging across models with uncorrelated errors reduces variance without increasing bias [18, 20]. The guarantee is conditional. When constituent models differ in accuracy, share correlated errors, or have heterogeneous biases, equal-weighted aggregation can produce an estimator no better than its best member. Whether these conditions hold in non-stationary financial time-series prediction is an empirical question that has received limited controlled study.

Many existing comparisons evaluate proposed ensembles against external baselines drawn from prior papers with different features, splits, or markets, making it difficult to attribute observed gains to the ensemble strategy rather than to differences in preprocessing. We close that gap with a controlled head-to-head comparison of Random Forest (RF) [3, 4, 18], Support Vector Regression (SVR) [1, 2, 10], Long Short-Term Memory (LSTM) [8, 9, 12, 13], and their equal-weighted average, under identical engineered features, identical chronological train/validation/test splits, identical scaling, and the Diebold and Mariano test [5] for statistical significance on the realized error sequences.

We make three contributions. (1) A controlled comparison of three model families and their equal-weighted ensemble on five sector-diversified U.S. assets across 2015–2024, with identical inputs and protocols. (2) Evaluation beyond point-estimate accuracy: error distribution, prediction bias, per-asset consistency, and pairwise Diebold and Mariano significance. (3) Empirical evidence that equal-weighted ensembling does not necessarily outperform its strongest constituent in this regime, and a bias/variance reading of why. The application is finance [11, 13, 17, 19], but the methodological claim concerns when ensemble aggregation is and is not a justified default.

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2 Background and Related Work

2.1 Predictability under Non-Stationarity

The difficulty of predicting financial time series is often framed through the Efficient Market [6] and Random Walk [7] hypotheses. The relevance for machine learning is practical rather than philosophical: low signal-to-noise ratios, weak autocorrelation, and non-stationary distributions [13, 15]. Models that excel on stationary benchmarks tend to generalize poorly across regimes, and gains observed in one period do not automatically transfer. Empirical work documents short-horizon predictability in the presence of frictions, behavioral effects, and information diffusion delays [11, 13, 17], motivating flexible models that do not require strong parametric assumptions.

2.2 Model Families

Random Forest [3, 4] aggregates decision trees on bootstrapped samples with random feature subsets, producing strong non-linear regressors that are robust to overfitting and easy to tune. It treats observations as exchangeable, so temporal structure must enter through engineered features, and it cannot extrapolate beyond the training-range of the target — relevant here because 2023–2024 prices exceed 2015–2020 levels for several assets. *Support Vector Regression* [1, 10] fits an ϵ -insensitive loss in an RBF-induced feature space; it is sensitive to bandwidth, regularisation, and ϵ , and is not range-bounded. *LSTM* [8, 12] addresses vanishing gradients in recurrent networks through gated memory cells, has many more parameters than RF or SVR, and benefits from longer training histories than are available at daily frequency for single equities; in our setup the LSTM is fed the same engineered features as the tabular learners, which substantially reduces the data-efficiency gap. Recent hybrid and attention-based architectures [16, 21] extend this family further, but our scope is restricted to the three canonical learners above.

2.3 Ensemble Assumptions and Comparison Gaps

The bias and variance decomposition (Equation (2)) makes three assumptions explicit: averaging helps when member variances are large, their pairwise covariances are small, and their biases are similar. Stacking trains a meta-learner that re-weights base predictions; boosting trains members sequentially to reduce residual error; gating allows weights to depend on observable context. Equal-weighted averaging makes none of these adjustments and is the right baseline for testing whether aggregation by itself adds value. Most prior ensemble studies in financial prediction compare proposed systems against external baselines drawn from different papers with different features, horizons, or markets [13, 17–20], and few report statistical significance over realized error sequences. Khattak *et al.* [11] observe that inconsistent baselines across studies undermine cross-study conclusions. Our protocol responds by fixing the feature pipeline, the splits, and the evaluation across all four predictors, and by applying the Diebold and Mariano test [5] to the realized squared-error sequences.

3 Methodology

3.1 Research Design

The experimental design isolates the effect of model choice (RF vs. SVR vs. LSTM vs. ensemble) from confounders due to features, splits, or preprocessing. *Controlled inputs*: all four predictors are trained and evaluated on the same engineered feature set with the same indicator definitions and transformations; the only representational difference is that the LSTM additionally receives temporal ordering through its sliding-window input. *Temporal splits*: the 2015–2024 history of each asset is split chronologically into training (2015–2020), validation (2021–2022), and test (2023–2024); all preprocessing statistics, scalar parameters, and rolling indicators are fit on training data only; the test set is used exactly once for the final evaluation, with no rolling refits or test-set hyperparameter selection. *Hyperparameter selection*: each model is tuned by grid search on validation RMSE with comparable search-space size, then retrained on the union of training and validation (or on training with validation used for early stopping) and evaluated once on the held-out test set. *Ensemble construction*: weights are fixed at 1/3 rather than tuned on validation, so the comparison isolates aggregation itself from the effects of a separately tuned combiner; a learned combiner would conflate whether aggregation helps with how its weights are obtained. *Scope*: the protocol does not compare engineered features against learned representations (the same engineered features go to every model), does not test multi-horizon performance (the target is fixed one day ahead), does not test learned combiners, and does not test economic value (the evaluation is in price units rather than returns or a backtested portfolio). Limiting the protocol in these dimensions makes any observed difference between the ensemble and its strongest constituent attributable to aggregation rather than to confounding choices in the surrounding pipeline.

3.2 Pipeline

The prediction system follows the sequential pipeline of Fig. 1. Daily OHLCV data are ingested per asset, augmented with technical indicators capturing trend, momentum, and volatility, scaled with statistics fit on training data only, and exposed to each base learner in the representation appropriate for its model family. The three base learners are trained in parallel and aggregated by a fixed equal-weighted average at test time. Evaluation runs once on a held-out test partition that the models do not see during tuning. Two principles govern the pipeline: strict temporal ordering throughout feature construction, scaling, and model fitting so no information from the test period leaks into training; and deliberate model-family diversity, so the ensemble assumption that members cover different regions of the function space is at least nominally satisfied.

3.3 Data and Features

Daily OHLCV bars for five sector-diversified U.S. assets retrieved from Yahoo Finance (AAPL, JPM, WMT, XOM, and the S&P 500 index GSPC), 1 January 2015 to 31 December 2024, about 2,520 trading days per asset. The chronological split is training 2015–2020, validation 2021–2022, test 2023–2024. From the raw bars we derive standard technical indicators [14]: SMA and EMA at several



Figure 1: End-to-end pipeline: OHLCV ingestion, technical-indicator feature engineering, chronological partitioning with training-only scaling, parallel training of Random Forest, SVR, and LSTM, equal-weighted ensemble formation, and held-out evaluation.

windows, RSI(14), MACD (line, signal, histogram), Bollinger Bands (upper, lower, percentage bandwidth), ATR(14), and five lagged closes. All rolling computations are forward-aligned, so the feature at time t uses only information up to t . MinMax scaling is fit on training data only and applied unchanged to validation and test.

3.4 Models

Let $X_t \in \mathbb{R}^d$ denote the feature vector at t and y_{t+1} the next-day close. RF (RandomForestRegressor) and SVR (RBF kernel) consume the flat X_t . The LSTM consumes sliding windows $\mathbf{X}_t = [X_{t-9}, \dots, X_t] \in \mathbb{R}^{10 \times d}$ and predicts y_{t+1} . Each model is tuned by grid search on validation RMSE with comparable search budgets: RF over $\{200, 500\}$ trees, depths $\{12, 20, \text{None}\}$, leaf sizes $\{1, 5\}$; SVR over $C \in \{1, 10, 100\}$, $\gamma \in \{0.01, 0.1, 1\}$, $\epsilon \in \{0.01, 0.1\}$; LSTM with one layer of 64 units, 10% dropout, Adam at learning rate 10^{-3} , batch size 64, early stopping (patience 20). Predictions are inverse-transformed to price units before any metric is computed.

3.5 Equal-Weighted Ensemble

With $\hat{y}_t^{RF}$, $\hat{y}_t^{SVR}$, $\hat{y}_t^{LSTM}$ the base predictions, the ensemble is

$$\hat{y}_t^{ENS} = \frac{1}{3}\hat{y}_t^{RF} + \frac{1}{3}\hat{y}_t^{SVR} + \frac{1}{3}\hat{y}_t^{LSTM}. \quad (1)$$

Weights are fixed at 1/3 rather than tuned, so any observed gap over the strongest member is attributable to aggregation itself rather than to a separately optimised combiner. Stacking, boosting, and regime-aware weighting are deferred to future work.

3.6 Evaluation and Significance

The primary metric is test-set RMSE. Complementary metrics cover accuracy (MAE, MAPE, MedianAE, R^2), error distribution (signed mean, std, IQR, P95 absolute error), and relative performance (NRMSE, MASE against a naive last-value baseline, Theil's U). For pairwise significance we apply the Diebold and Mariano test [5] on squared-error losses, using a Newey–West variance estimator to account for serial correlation in forecast errors. The bias and variance decomposition for an equal-weighted average of n models,

$$\mathbb{E}[(\bar{f} - y)^2] = \bar{b}^2 + \frac{1}{n^2} \sum_i \text{Var}(f_i) + \frac{1}{n^2} \sum_{i \neq j} \text{Cov}(f_i, f_j), \quad (2)$$

provides the structural lens for interpreting the results.

Table 1: Regression metrics on the 2023–2024 test set (lower is better, except R^2).

Metric	LSTM	SVR	Ens.	RF
RMSE	17.70	17.78	18.13	20.33
MAE	7.01	7.07	7.29	8.45
MAPE (%)	0.89	0.90	0.92	1.07
MedianAE	1.22	1.23	1.29	1.50
Error mean	+0.12	+1.16	+1.78	+4.11
Error std.	17.71	17.74	18.05	19.91
Error IQR	2.43	2.45	2.53	2.90
P95 abs. err.	43.05	45.47	45.29	52.47
NRMSE	0.00923	0.00930	0.00945	0.01063
MASE	0.81	0.75	0.84	0.89
Theil's U	0.00804	0.00810	0.00823	0.00926

4 Results

4.1 Overall Accuracy

Table 1 summarises test-set metrics across all five assets. Except for MASE (see below), every row pools the test residuals of all five assets into a single error vector and computes the metric once on the pool; the result is therefore dominated by the S&P 500 index, whose price level is roughly 20–30 times those of the individual equities, so its absolute errors dwarf the others. This pooled convention is why the overall RMSE (17.70 for LSTM) is much larger than the unweighted mean of the per-asset RMSEs in Table 3 (9.38 for LSTM): the two numbers answer different questions and are not directly comparable. On the pooled metric LSTM has the lowest RMSE (17.70) and MAE (7.01); SVR is close behind (17.78 / 7.07); the equal-weighted ensemble is third (18.13 / 7.29); RF last (20.33 / 8.45). R^2 saturates near 0.9999 across predictors (absolute-price target is trend-dominated), so the predictors separate on RMSE, MAE, and bias rather than on explained variance. MASE is the exception to the pooling convention: it is computed per asset against that asset's own in-sample naive one-step baseline and then averaged across the five assets, so it carries a separate scaling denominator per asset. This is why its ranking need not match the pooled MAE row: SVR has the lowest aggregate MASE (0.75) despite a slightly higher pooled MAE than LSTM, because it is relatively stronger on the assets whose naive baseline is harder to beat, whereas the pooled MAE is governed by the index.

4.2 Statistical Significance

Diebold and Mariano p -values for pairwise RMSE comparisons (Table 2) show LSTM and SVR are statistically tied ($p = 0.494$); LSTM is ahead of the ensemble at the conventional level ($p = 0.035$); SVR beats the ensemble more strongly ($p = 0.001$); all three beat RF ($p < 0.001$). With six pairwise comparisons the Bonferroni threshold is $\alpha/6 \approx 0.0083$ at $\alpha = 0.05$. Under this correction the SVR–ensemble comparison and all three comparisons against RF remain significant, but the LSTM–ensemble comparison (Bonferroni-adjusted $p \approx 0.21$) does not; we therefore treat LSTM's edge over the ensemble as marginal and rest the negative result on the SVR–ensemble gap, which survives correction. The LSTM–SVR tie reflects similar realized losses; on its own the DM test compares

Table 2: Diebold and Mariano p -values for pairwise RMSE comparisons.

	LSTM	SVR	Ens.	RF
LSTM	—	0.494	0.035	< 0.001
SVR	0.494	—	0.001	< 0.001
Ensemble	0.035	0.001	—	< 0.001
RF	< 0.001	< 0.001	< 0.001	—

Table 3: Per-asset RMSE on each asset’s own error sequence (lowest per row in bold). The Mean row is the unweighted average of the five per-asset values and is not comparable to the pooled overall RMSE of Table 1.

Asset	LSTM	SVR	Ens.	RF
AAPL	2.58	2.62	2.69	3.03
JPM	2.60	2.59	2.64	2.81
WMT	0.69	0.70	0.73	0.82
XOM	1.60	1.56	1.56	1.81
GSPC	39.41	39.58	40.36	45.26
Mean	9.38	9.41	9.60	10.75

loss sequences and does not establish that the two forecasts are correlated, so we measure that separately below.

The high-covariance condition can be read directly off Table 1. Because the ensemble error is the equal-weighted average of the three constituent errors, their variances satisfy $9\sigma_{ENS}^2 = \sum_i \sigma_i^2 + 2\sum_{i<j} \text{Cov}(e_i, e_j)$, where the σ_i are the error standard-deviation rows of Table 1. Substituting $\sigma_{RF} = 19.91$, $\sigma_{SVR} = 17.74$, $\sigma_{LSTM} = 17.71$, and $\sigma_{ENS} = 18.05$ gives $\sum_{i<j} \text{Cov}(e_i, e_j) \approx 954$, which corresponds to an average pairwise error correlation of approximately 0.94. The three forecast-error sequences are thus strongly correlated, so by Equation (2) the variance-reduction term available to any equal-weighted average over them is small.

4.3 Per-Asset Consistency

Per-asset RMSE (Table 3) shows SVR is best on AAPL, JPM, and XOM; LSTM is best on WMT and the S&P 500 index; the equal-weighted ensemble lies between the best and worst individual learner on every asset; and RF is last on every asset. The aggregate ranking holds uniformly across sectors and is not driven by a single asset. The alternation between LSTM and SVR as best-per-asset implies that the two strongest predictors are not strict substitutes; a learned combiner with context-dependent weights could exploit this, and equal-weighted averaging does not.

4.4 Bias and Spread

The error-distribution rows of Table 1 show that LSTM has the tightest dispersion (IQR 2.43, P95 absolute error 43.05); the ensemble is wider than either strong member (IQR 2.53); RF is widest (IQR 2.90). Signed mean errors of +0.12 (LSTM), +1.16 (SVR), +1.78 (ensemble), and +4.11 (RF) show systematic under-prediction across all four, most severely for RF. The ensemble bias of +1.78 lies between those of its members rather than below them, consistent with simple

averaging of heterogeneously biased predictors. Inflated bias plus unimproved spread explains the RMSE ranking. The roughly 15% RMSE gap between LSTM and RF is the largest pairwise difference among the four predictors; the corresponding gap between LSTM and the ensemble is smaller in absolute terms (about 2.4%) but is systematic across metrics. On every accuracy metric in Table 1, the ensemble lies between its strongest and weakest constituent, which is the expected behaviour of simple averaging when constituent quality is unequal.

4.5 Theoretical Consistency Check

The empirical ranking matches the structural prediction of Equation (2). The signed mean errors range over more than four units of price, indicating heterogeneous biases across predictors. The average pairwise error correlation of approximately 0.94, recovered from the variance decomposition in Section 4 rather than inferred from the DM test, confirms the high-covariance condition. Together these properties limit the gain available from equal-weighted averaging: the squared-bias term of the ensemble is closer to a blend of large and small biases than to the smallest individual bias, and the variance term is reduced by far less than $1/n$ because of inter-model covariance. The observed ordering $\text{LSTM} \approx \text{SVR} < \text{ensemble} < \text{RF}$ is therefore the ordering Equation (2) predicts under exactly these conditions, which makes the result structural rather than incidental to the specific assets or features used here.

5 Discussion

Three properties of the constituent models break the optimistic case for equal-weighted averaging in this setting. First, accuracy is heterogeneous: RF RMSE (20.33) is roughly 15% above LSTM’s (17.70), so averaging drags the ensemble away from the best member. Second, the three forecast-error sequences are strongly correlated: the variance decomposition applied to the error standard deviations of Table 1 implies an average pairwise correlation of approximately 0.94, and high covariance limits the variance-reduction term in Equation (2). Third, the four predictors have visibly heterogeneous biases (+0.12 to +4.11), so the squared mean bias of the average is not necessarily smaller than that of the best member. The observed ranking $\text{LSTM} \approx \text{SVR} < \text{ensemble} < \text{RF}$ is the ordering Equation (2) predicts under exactly these conditions, which provides a sanity check that the result is structural rather than incidental to specific assets or features.

The practical implication is methodological. In tasks with non-stationary, low-signal data and heterogeneous candidate models, ensemble construction should be approached as a model-selection problem rather than as a default step. Three cheap diagnostics on validation data — per-model RMSE, pairwise error correlations, and per-model signed mean error — directly target the assumptions in Equation (2). When they indicate unequal accuracy, correlated errors, or heterogeneous bias, equal-weighted averaging is unlikely to outperform careful individual model selection, and a learned combiner (stacking, gating, or regime-aware weighting) is the natural next step. The negative result here does not contradict prior findings on the value of ensembles in general [13, 15, 18, 20]; it qualifies the assumption that aggregation is automatically beneficial.

The per-asset table carries a secondary message: SVR is best on AAPL, JPM, and XOM; LSTM is best on WMT and the S&P 500 index. The two strongest predictors differ on which assets they handle best, even though their aggregate accuracy is statistically tied. A combiner with weights that depend on the asset or on observable context such as recent realized volatility could exploit this structure, but equal-weighted averaging does not. The fact that aggregation underperforms its best member in our experiments does not imply that aggregation is uninformative; it implies that the chosen aggregation rule does not exploit the structure of disagreement among members. A more sophisticated combiner is the natural object of comparison once equal weighting has been ruled out as a sufficient strategy.

Three over-readings of the headline finding are worth ruling out. The result does not say that LSTM is the best machine-learning model for stock prediction in general: under a different feature set, asset universe, or horizon, the ranking can differ, and our LSTM is statistically tied with SVR even within the present setting. The result does not say that ensembles do not work: it says that the simplest aggregation rule, equal-weighted averaging across a tabular and a sequential predictor with comparable accuracy and a weaker tabular predictor, did not beat the best member here. The result does not translate to claims about trading profitability or economic value: the evaluation is in price units, and a model that wins on RMSE does not necessarily generate profits net of transaction costs and slippage.

6 Limitations

The findings are bounded by the scope of the experimental setting. The evaluation covers five U.S. assets across 2015–2024 at daily frequency; generalization to other markets, asset classes, frequencies, and time periods is not tested. The features are technical indicators computed from OHLCV data only; adding fundamentals, news, sentiment, or limit-order-book microstructure may alter relative model performance, and the relative ranking of model families is not guaranteed to be invariant to feature set. The ensemble strategy studied is equal-weighted averaging, not stacking, boosting, gating, or regime-aware weighting; conclusions about ensemble performance are therefore specific to this aggregation rule and do not generalise to all ensemble strategies. The forecast horizon is one day ahead. The evaluation is predictive and statistical: we report regression accuracy and the Diebold and Mariano test on squared errors, but we do not assess trading profitability, transaction costs, slippage, risk-adjusted returns, or portfolio behavior. The reported numbers must not be interpreted as financial advice or as evidence of economic value. The pairwise Diebold and Mariano p -values in Table 2 are uncorrected for multiple comparisons. Under Bonferroni correction across the six pairwise tests at $\alpha = 0.05$, the SVR–ensemble comparison and all comparisons involving RF remain significant, but LSTM’s edge over the ensemble ($p = 0.035$) becomes marginal; the negative result therefore rests on the SVR–ensemble gap rather than on the LSTM–ensemble gap.

7 Future Work and Conclusion

Several extensions follow naturally from the analysis. First, learned combiners (stacking with a regularised meta-learner, gradient-boosted

residual correction, or constrained least-squares weights tuned on validation) can be compared against equal weights under the same protocol used here; the experimental machinery is already in place. Second, adaptive weighting that responds to recent error or to detected regimes may reduce the effective bias gap among members and is a natural response to non-stationarity [9, 11]. Third, multi-horizon evaluation and uncertainty quantification (prediction intervals, calibration) would extend the evaluation beyond point estimates and are arguably more informative for downstream decision-making. Fourth, alternative data sources (fundamentals, macroeconomic releases, news, social media, limit-order-book microstructure) may change the dependence structure among models and shift the cost–benefit tradeoff of aggregation. Finally, a complementary economic evaluation incorporating transaction costs and portfolio constraints is needed to translate predictive accuracy into operational value; the present paper does not undertake it.

Within the scope above, the equal-weighted ensemble did not outperform the strongest individual models: LSTM achieved the lowest pooled RMSE (17.70), SVR was statistically tied with LSTM ($p = 0.494$), the ensemble was third (SVR ahead of it at $p = 0.001$, surviving Bonferroni correction; LSTM ahead at a marginal $p = 0.035$), and RF was last. The bias and variance decomposition explains why under the observed accuracy heterogeneity, error correlation, and bias dispersion. The methodological lesson generalises beyond financial prediction: in any non-stationary regression setting where several heterogeneous model families are candidates, ensemble construction should be approached as a model-selection problem rather than as a default step.

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